

L_p -NORM ESTIMATION OF THE LOCATION PARAMETER OF A SYMMETRIC DISTRIBUTION

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SUMMARY

The estimation of the local parameter θ is considered by minimizing $\sum_{i=1}^n |x_i - \theta|^p$ where p is made to be some function of the kurtosis of the underlying data set which is assumed symmetrical, but is in general non-normal. These estimates of the location parameter are compared to "robust" estimators which are conventionally used when data deviates from normality such as the Huber, Hampel and trimmed mean estimators. The mean square error of the estimators is used as a yardstick of comparison. Over the range of different distributions and sample sizes considered it is shown that the method proposed provided superior estimates to the alternatives considered.

1. Introduction

An impressive array of robust estimators of location parameters has been proposed in the literature. These estimators are primarily constructed so as to have superior statistical properties to the least squares estimator $\bar{X}$ when the underlying distribution deviates from normality. In particular, these estimators usually exhibit the characteristic of being less sensitive than $\bar{X}$ to outliers or bad data points. In other words, these estimators are usually constructed to be "better" (according to some criterion) than $\bar{X}$ when the underlying distribution is leptokurtic with respect to the normal. Less attention has been given to the performance of estimators which outperform $\bar{X}$ when the underlying distribution is either platykurtic or leptokurtic with respect to the normal.

The purpose of this paper is to propose a new robust method for location parameter estimation and to evaluate the performance of the proposed estimator against a selection

of commonly used alternatives.

2. Previous estimators proposed

In the statistical literature numerous population parameters have been proposed as a measure of the location of the distribution. In particular, the mean ($E(X)$), population median and population midrange have received special attention. Hence in estimating the location parameter the first problem is to establish what one considers to be the most appropriate measure of location for the particular data set under consideration. Once the appropriate measure has been chosen the problem of estimation can then be tackled.

To avoid this subjective choice of the most suitable location parameter, consideration was limited to symmetric populations so that

$$E(X) = \theta_{\text{median}} = \theta_{\text{midrange}} = \theta$$

Two closely related families of estimators have been given attention in the literature, namely the class known as *M-estimators* and the class known as *L-estimators*.

The *M-estimators* consider the general problem of finding the value of θ which minimizes

$$(2.1) \quad \sum_{i=1}^n \rho(x_i - \theta)$$

for a random sample $x_1, x_2, \dots, x_n$ from the underlying distribution with probability density function $f(x - \theta)$ where θ is the location parameter and ρ is some non-constant function. Equation (2.1) will be minimized when

$$(2.2) \quad \sum_{i=1}^n \psi(x_i - \theta) = 0$$

where $\psi = \rho'$ (the first derivative of ρ with respect to θ).

A number of ψ functions have been proposed and the reader is referred to Andrews et al (1972) for a complete discussion of these estimators. The following ψ functions were considered in this study.

The original M-estimator was that due to Huber (Andrews et al (1972)) with

$$(2.3) \quad \psi(y) = \begin{cases} -a & ; & y < -a \\ y & ; & -a \leq y \leq a \\ a & ; & y > a \end{cases}$$

In practice a scale invariant form of the function is normally used, with θ being the solu-

tion of

$$\sum_{i=1}^n \psi\left(\frac{x_i - \theta}{d}\right) = 0$$

where d is some robust estimate of the spread of X . Hogg (1977) cites the use of

$$(2.4) \quad d = \text{median}_i |x_i - \text{median}_i(x_i)| / (0.6745).$$

(The factor 0.6745 makes this an approximately unbiased estimate of scale when the error model is Gaussian. See Holland and Welsch (1977).)

An extension of Huber's idea is one due to Hampel (1974) where

$$(2.5) \quad \psi(x; a, b, c) = (\text{sign } x) \begin{cases} |x| & ; & 0 \leq |x| < a ; \\ a & ; & a \leq |x| < b ; \\ \frac{c-|x|}{c-b}a & ; & b \leq |x| < c ; \\ 0 & ; & c \leq |x| \end{cases}$$

for some predetermined a , b and c .

Andrews (1972) has suggested

$$(2.6) \quad \psi(x) = \begin{cases} \sin(x/d), & |x| < d\pi \\ 0, & \text{otherwise} \end{cases}$$

for some predetermined d .

The *L-estimators* (as before see Andrews et al (1972) for a full discussion) have received considerable attention, most particularly the α -trimmed mean,

$$(2.7) \quad m(\alpha) = \frac{1}{h} \sum_{i=g+1}^{n-g} X_{(i)},$$

where α is selected so that $g = n\alpha \in \mathbb{Z}$, the set of positive integers, $h = n - 2g$, and $X_{(i)}$ is the i th ordered value

Hogg (1967) has proposed the use of $m(\alpha)$, with α a function of the sample kurtosis (k) in a stepwise continuous way.

He proposes

$$(2.8) \quad \begin{cases} m^c(1/4), & \hat{k} < 2,0 \\ m(0), & 2,0 \leq \hat{k} \leq 4,0 \\ m(1/4), & 4,0 < \hat{k} \leq 5,5 \\ m(1/2), & \hat{k} > 5,5 \end{cases}$$

($m^c(1/4)$ is the mean of the trimmings of $m(1/4)$).

Numerous other estimators have been proposed. For example Hogg (1974) proposed the Q-statistic as an alternative to kurtosis as a measure of tail stretch and then proposed a modification of (2.8) based on this statistic. In a similar vein Prescott (1978), and de Wet and van Wyk (1979), have proposed adaptive trimmed mean estimators based on the Q-statistic.

3. New Estimator Proposed

Reconsideration of the M-estimator approach shows that minimization of

- (a) $\sum_{i=1}^n |x_i - \theta|$ yields $\hat{\theta} = X_{\text{median}}$ (the sample median and maximum likelihood estimate of θ for the Laplace distribution).
- (b) $\sum_{i=1}^n (x_i - \theta)^2$ yields $\hat{\theta} = \bar{X}$ (the maximum likelihood estimate of θ for the normal distribution).
- (c) $\sum_{i=1}^n |x_i - \theta|^\infty$ yields $\hat{\theta} = X_{\text{midrange}}$ (the sample midrange and maximum likelihood estimate of θ for the uniform distribution).

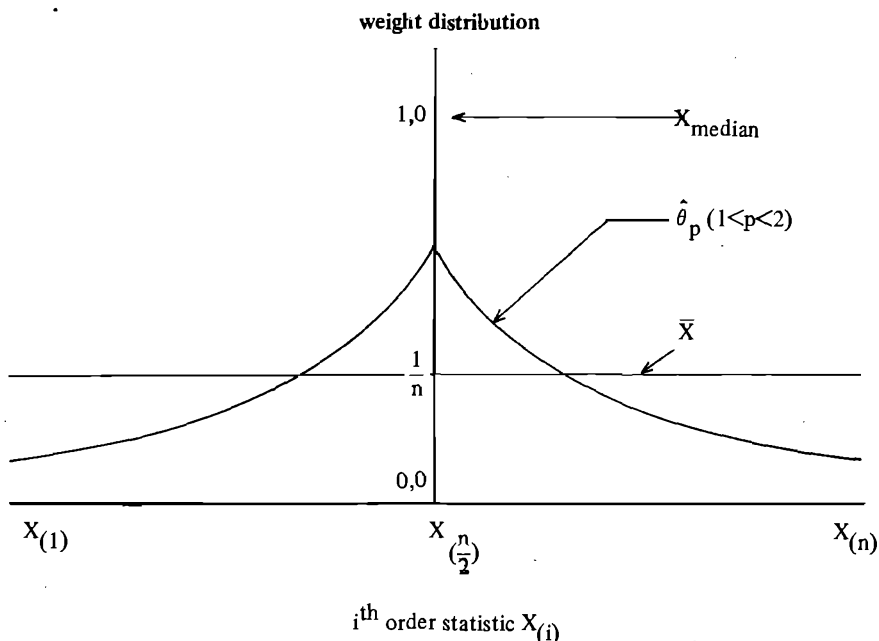
If we consider the minimization of

$$(3.1) \quad \sum_{i=1}^n |x_i - \theta|^p$$

where p is a continuous function of tail stretch, it would appear that one may be able, at least in terms of maximizing minimum efficiency, to improve on the blanket use of either $\bar{X}$ or X_{median} and even the use of the Huber and Hogg estimators.

It is clear that using a value of $p = 2$ assigns equal weight to all observations in the calculation of $\hat{\theta}$; use of $p = 1$ assigns all the weight to the middle observation (in the case of n odd) and none to the others; use of $p = \infty$ assigns half weight to each of the end observations and none to the other. Use of $1 < p < \infty$ therefore assigns weight to the observations; the closer p is to 1 the more the weights shift towards the central values (see Figure 1).

FIGURE 1



Such a procedure may have more intuitive appeal than the truncated mean or Huber approach, because in those procedures information regarding the magnitude of data points falling in the tails is lost in the sense that the influence curves (Hampel (1974)) of such estimators are constant (non zero) past the relevant cut off points. Thus although the existence of outliers influence such estimators and in fact as Hampel notes for the *trimmed mean*, exert “the maximum possible influence on each side” the actual values of such outliers in the tails are irrelevant for a given trimming proportion.

4. Selection of a Suitable p

A formula based on a simulation study of the robust estimation of the parameters of a regression model with a symmetrical error distribution has been proposed by Money et al (1981). When the kurtosis is known this has been shown to be superior to either L_2 or L_1 estimation in terms of maximin efficiency based on empirical generalised variance (over a range of distributions). This formula

$$(4.1) \quad p = 1 + \frac{9}{k^2}$$

proposed the use of least squares under a normal distribution, and a continuous decrease in p for increasing k with a limiting value of $p = 1$.

In the general case, when k is unknown, use of the sample estimate of k , $\hat{k}$ instead of k produced parallel comparative advantages over L_2 and L_1 regression (Money et al (1981)).

Estimation of the location parameter is merely a special case of the regression problem and hence the above procedure could be used. This would lead to the following procedure.

The L_p Method of Estimating the Location Parameter of a Symmetric Distribution.

$$\text{Minimize } \sum_{i=1}^n |x_i - \theta|^p$$

$$\text{where } p = 1 + \frac{9}{\hat{k}^2}$$

$$\text{and } \hat{k} = 3 + \frac{k_4}{k_2^2}$$

k_i being the unbiased estimate of the i th cumulant.

As previously mentioned, various alternative measures of tail stretch have been proposed in the literature, most notably Hogg's Q-statistic. It does, however, appear from the literature that none have any clear cut advantage over kurtosis. Therefore while p could be expressed as a function of Q this was not done as it is felt that it would not lead to significantly improved results.

5. Design of the Simulation

A variety of statistical distributions were used to generate the raw data sets. All were symmetric and were chosen to cover a wide range of kurtosis. The sample sizes n (for each iteration of the simulation) used were chosen as 10, 30 and 50; and 500 iterations were performed for each sample size and each distribution. The basic criterion for evaluation of each estimator was sample MSE (the sample mean of the 500 squared deviations of the calculated values from the true value). Efficiencies (relative to the best estimator for each n and distribution) then can be calculated and the estimator may be evaluated according to its "best-worst-performance" of minimax efficiency.

Several distinct methods were used in the calculation of the location parameter θ .

- A. Least absolute deviation ($\hat{\theta}$ = sample median)
- B. The adaptive L_p -method
- C. Ordinary least squares ($\hat{\theta} = \bar{X}$).

- D. Huber method* ($a = 1,5$)⁺
- E. Hampel method* ($a = 1,7$; $b = 3,4$ and $c = 8,5$)⁺
- F. Andrews method* ($d = 2,1$)⁺
- G. Trimmed mean (using Hogg's criterion (2.8)).

The algorithm used for D, E and F was that due to Huber (1975) with the median used as a starting value and all the programs were written in UNIVAC (1100 series) DOUBLE PRECISION ASCII FORTRAN. Data sets were simulated from:

- (a) Uniform distribution (kurtosis = 1,8).
- (b) Normal distribution (kurtosis = 3,0).
- (c) Contaminated normal distributions (kurtosis = 3,5; 4,0; 4,5; 5,0 and 5,5).
- (d) Laplace distribution (kurtosis = 6,0).
- (e) Cauchy distribution (kurtosis undefined).

Parameters were chosen so that each distribution (apart from the Cauchy) had $\theta = 0$ and $\sigma^2 = 9$. [The Cauchy distribution with parameters α and β has no moments but is symmetric about α which was chosen to be zero. β was determined by specifying that the 95th percentile of the Cauchy distribution had to coincide with the 95th percentile of the Normal (0,9) distribution.]

6. Experimental Results

The aim of the simulation is to gain insight into the use of the new proposed L_p -estimators of θ vis-a-vis the use of some of the more conventional robust estimators.

All the estimators considered, with the possible exception of B, are well known to be unbiased. Harvey (1978) has argued that the L_p -norm estimates are unbiased if the first moment exists and the underlying distribution is symmetric. By considering sample MSE any conjecture on this point is avoided as this statistic gives us a joint measure of bias and variance.

* D, E, and F were all applied using the scale invariant form due to Hampel (Hogg (1974)).

+ See Hogg (1974)

TABLE 6.1
Sample Mean Square Error of the Location Parameter Estimates
N = 10 (500 iterations)

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	2,134	0,810	0,904	1,200	1,154	1,093	0,891
Normal	3,0	1,168	0,865	0,798	0,862	0,857	0,834	0,876
Con. Normal	3,5	1,172	1,076	1,044	1,016	1,014	1,016	1,102
Con. Normal	4,0	1,051	0,954	0,938	0,917	0,921	0,916	0,966
Con. Normal	4,5	0,792	0,851	0,884	0,753	0,770	0,782	0,859
Con. Normal	5,0	0,616	0,827	0,910	0,693	0,716	0,737	0,840
Con. Normal	5,5	0,336	0,691	0,783	0,524	0,560	0,600	0,676
Laplace	6,0	0,710	0,869	1,005	0,769	0,783	0,800	0,859
Cauchy	undefined	0,068	0,141	10,270	0,120	0,107	0,122	0,116
Average Sample Mean Square Error		0,894	0,787	1,948	0,762	0,765	0,767	0,798

TABLE 6.2
Efficiency of Estimates (based on M.S.E.)
N = 10

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	38	100	90	68	70	74	91
Normal	3,0	68	92	100	93	93	96	91
Con. Normal	3,5	87	94	97	100	100	100	92
Con. Normal	4,0	87	96	98	100	99	100	95
Con. Normal	4,5	95	88	85	100	98	96	88
Con. Normal	5,0	100	74	68	89	86	84	73
Con. Normal	5,5	100	49	43	64	60	56	50
Laplace	6,0	100	82	71	92	91	89	83
Cauchy	—	100	48	1	57	64	56	59
Minimum Efficiency		38	48	1	57	60	56	50

TABLE 6.3
Sample Mean Square Error of the Location Parameter Estimates
N = 30 (500 iterations)

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	0,798	0,200	0,312	0,355	0,333	0,337	0,254
Normal	3,0	0,457	0,313	0,297	0,318	0,312	0,304	0,314
Con. Normal	3,5	0,396	0,303	0,294	0,294	0,295	0,289	0,291
Con. Normal	4,0	0,354	0,310	0,304	0,296	0,297	0,294	0,315
Con. Normal	4,5	0,283	0,297	0,331	0,275	0,283	0,290	0,300
Con. Normal	5,0	0,205	0,227	0,285	0,214	0,225	0,236	0,236
Con. Normal	5,5	0,112	0,160	0,284	0,169	0,179	0,197	0,165
Laplace	6,0	0,183	0,227	0,304	0,230	0,237	0,243	0,236
Cauchy	Undefined	0,021	0,024	2,590	0,033	0,029	0,032	0,023
Average Sample Mean Square Error		0,313	0,229	0,556	0,243	0,243	0,247	0,237

TABLE 6.4
Efficiency of Estimates (based on M.S.E.)
N = 30

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	25	100	64	56	60	59	79
Normal	3,0	65	95	100	93	95	98	95
Con. Normal	3,5	73	95	98	98	98	100	99
Con. Normal	4,0	83	95	97	99	99	100	93
Con. Normal	4,5	97	93	83	100	97	95	92
Con. Normal	5,0	100	90	72	96	91	87	87
Con. Normal	5,5	100	70	39	66	63	57	68
Laplace	6,0	100	81	60	80	77	75	78
Cauchy	—	100	88	1	64	72	66	91
Minimum Efficiency		25	70	1	56	60	57	68

TABLE 6.5
Sample Mean Square Error of the Location Parameter Estimates
N = 50 (500 iterations)

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	0,487	0,096	0,173	0,185	0,176	0,184	0,125
Normal	3,0	0,286	0,180	0,175	0,182	0,179	0,178	0,183
Con. Normal	3,5	0,241	0,183	0,185	0,179	0,179	0,179	0,186
Con. Normal	4,0	0,196	0,166	0,183	0,164	0,166	0,168	0,170
Con. Normal	4,5	0,148	0,144	0,177	0,136	0,141	0,148	0,148
Con. Normal	5,0	0,110	0,122	0,173	0,118	0,126	0,135	0,135
Con. Normal	5,5	0,075	0,095	0,211	0,108	0,114	0,132	0,086
Laplace	6,0	0,104	0,120	0,183	0,133	0,137	0,142	0,126
Cauchy	undefined	0,011	0,011	1,618	0,016	0,019	0,017	0,011
Average Sample Mean Square Error		0,184	0,124	0,342	0,136	0,137	0,143	0,130

TABLE 6.6
Efficiency of Estimates (based on M.S.E.)
N = 50

Distbn.	Kurtosis	A	B	C	D	E	F	G
Uniform	1,8	20	100	55	52	55	52	77
Normal	3,0	61	97	100	96	98	98	96
Con. Normal	3,5	74	98	97	100	100	100	96
Con. Normal	4,0	84	99	90	100	99	98	96
Con. Normal	4,5	92	94	77	100	96	92	92
Con. Normal	5,0	100	90	64	93	87	81	81
Con. Normal	5,5	100	79	36	69	66	57	87
Laplace	6,0	100	87	57	78	76	73	83
Cauchy	—	100	100	1	69	58	65	100
Minimum Efficiency		20	79	1	52	55	52	77

Examination of Tables 6.1 and 6.2 reveals that for a sample size of 10, B is marginally outperformed in terms of a maximin efficiency criterion by D, E, F and G. Using an average sample MSE criterion, B is outperformed by D, E and F. For sample sizes of 30 and 50, (Tables 6.3 to 6.6), B is the best performer in terms of a maximin efficiency criterion and an average sample MSE criterion.

It is seen that the relative superiority of B increases as the sample size increases. Tukey has commented (Andrews et al (1972)) that adaptive estimators "*come into their own for fairly large samples, probably beyond 50*" and although the point is taken, it is seen that B's performance for n is 30 is still excellent and for n is 10, at worst, mediocre.

7. Conclusions

It is seen that B does very well in this study in comparison to the class of estimators studied. It performs adequately for sample size equal 10 and extremely well for sample size 30 and 50.

It should be noted in this study that all the non-normal distributions considered, except one, (the uniform) had kurtosis greater than 3.0. Thus possibly disproportionate weight has been given in this study to long tailed distributions. (This was in fact a criticism of the Princeton study — see Wegman and Carroll (1977).) If this had not been the case, one could assume from the above results that the relative performance of B would have been even better (since B is better than any alternatives for the uniform distribution in all three sample sizes examined.)

It is also worth noting that G probably does best of the alternatives to B. G is however much easier to compute and may well provide a viable alternative to B if only limited computational facilities are available.

It can be argued therefore, on the grounds of the results obtained, that for the distributions considered, the adaptive L_p -norm methods are superior to blanket use of any of the alternative schemes for the estimation of the location parameter except in the case when the sample is very small.

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